A History of Rationality in Economics: a Computational Approach

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Abstract
This paper offers a concrete, method-driven account of how quantitative techniques can illuminate the history of a capacious concept in economics—rationality. We assemble a large full-text corpus (nearly 300,000 economics articles since 1900) and pair it with structured citation data. On the textual side, we use large language models (LLMs) to trace semantic shifts over more than a century and to identify a corpus of texts centered on rationality, which we then analyze with bibliometrics and network methods. Our objectives are twofold: first, to provide a concrete methodological account that foregrounds the practical choices—from data collection to interpretation—that shape corpus construction and the reading of results; second, to show that unsupervised models, when combined with robustness checks and close reading, can serve as powerful tools for both confirmation and discovery in the history of economics. To ensure transparency and reuse, we release a Shiny application that exposes the indicators we rely on for qualitative interpretation and makes the workflow auditable and replicable.

KEYWORDS
rationality; economics; bibliometrics; text mining; natural language processing; large language models

JEL CLASSIFICATION
B00; B20

> **Warning**
>
> This is a work in progress. The discussion Section in particular will be expanded in future versions.

> **Interactive application to explore the data**
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> We have developed an interactive application to explore the data and results presented in this paper. It is available at: https://0199205a-7b17-ab11-595d-8f12aae160dd.share.connect.posit.cloud. The first tab allows to explore the bibliometric networks; the second tab is a global summary of textual clusters.

Introduction

In the last decade, the number of history of economics papers employing quantitative methods has increased (see e.g., Goutsmedt, Claveau, and Herfeld 2023 special issue). Several essays have adopted a reflexive stance, examining the implications of using quantitative methods in the history of economics (Cherrier and Svorenčík 2018; Edwards, Giraud, and Schinckus 2018). These contributions have aimed to provide broad discussions on the use of such methods. However, given the virtually unlimited variety of quantitative approaches potentially relevant to historians of economics—and the wide array of research questions they can address—these reflections often remain abstract and offer little practical guidance. As a result, they tend to provide useful broad overviews, but without clearly articulating what quantification entails in practice, and how it can be both meaningful and challenging to integrate into historical research.

Our contribution sits between a historical study that uses quantitative methods to answer a specific question and a general methodological discussion of those methods. From the collection of data to the interpretation of results, we illustrate concretely how these methods are useful and examine, in practice, the methodological choices they entail. We present a step-by-step application of specific quantitative methods to a broad topic: the history of rationality in the 20th century.

Our article uses two types of data: textual and bibliometric. Beyond basic statistics (such as counting words or citations), we rely on unsupervised models to classify our corpus into categories—what we call "textual clusters" and "bibliometric communities."[1] These methods depart from standard econometric regressions. Their goal is not necessarily to provide a "measure" (Grimmer, Roberts, and Stewart 2022)—though they can be adapted to do so—but to organise in categories large corpora and enable accelerated and "distant reading" (Moretti 2013; Guldi 2023). They are well suited to historical inquiry: when incorporating time explicitly, they allow researchers to map the discipline at different points of time, to identify the emergence and decline of subjects and concepts, and to assess the influence of specific economists.

Above all, our discussion aims to illustrate a core principle for historical inquiry with such unsupervised quantitative methods. They require continuous back-and-forth between aggregate quantitative results, complementary indicators used for interpretation, and close reading of primary sources. These methods function as "discovery methods" (Grimmer, Roberts, and Stewart 2022): they facilitate the exploration of large datasets to reveal historical patterns. They often confirm, and sometimes complement, established findings. They also reveal pitfalls and blind spots in existing research.

The idea of "rationality" is an effective focus for a concrete demonstration. First, many historians of economics engage with it in one way or another, which makes the exercise relevant for a broad audience. Second, because the concept is broad and pervasive in economics, applying quantitative methods to a very large corpus is particularly informative.[2] We use a corpus of around 300 000 articles extending back to 1900 to show how these methods can handle long time horizons. Third, the multiple meanings attached to "rationality" show how textual methods can help capture this semantic plurality.

[1] The bibliometric communities, identified through network analysis, could also be called clusters. But we opted for "communities" in order to distinguish them from the "textual clusters".

[2] To be clear, we don't think that quantitative methods are only helpful for very large corpora. However, it is where their surplus value may appear as the more obvious.

Some milestones in the history of rationality in economics

Rationality is one of the most central and encompassing concepts in economics, alongside "market" and "competition." Several works trace its evolution across decades or within subfields, and a substantial literature examines specific components such as expected utility theory, rational expectations, and bounded rationality (e.g., Moscati 2016; Pedro G. Duarte 2025; E.-M. Sent 2008). Our aim here is not to offer an exhaustive survey but to pinpoint key developments and debates in this long and complex history.

The idea that agents behave rationally is implicit in classical arguments, for example in Ricardian rent theory, where farmers allocate land by marginal fertility and profitability. While a single origin is unlikely, the concept likely gained prominence with the rise of scientific and abstract reasoning in the late nineteenth century. The scope of rational behavior was closely tied to debates on the proper domain of economic analysis (Giocoli 2003). A well-known illustration is John Stuart Mill's *Essays on Some Unsettled Questions of Political Economy*:

> What is now commonly understood by the term "Political Economy" is not the science of speculative politics, but a branch of that science. It does not treat of the whole of man's nature as modified by the social state, nor of the whole conduct of man in society. It is concerned with him solely as a being who desires to possess wealth, and who is capable of judging of the comparative efficacy of means for obtaining that end. (Mill 1844)

With the marginalists, the economic agent was increasingly defined by calculative capacity. Jevons portrayed the agent as a "calculating machine" balancing pleasures and pains, a view later formalized as utility maximization (Maas 1999). From marginalist theory emerged a first meaning of rationality: utility maximization under constraints, with rational behavior understood as the consistent selection of means that maximize utility. This conception became integral to the development of neoclassical demand theory, culminating in John R. Hicks's *Value and Capital* (1939).

The Second World War and the Cold War marked a turning point. Key sites included RAND and the Cowles Commission, where operations research and linear programming took shape; figures such as Jacob Marschak, Tjalling Koopmans, and George Dantzig were central (Mirowski 2002; Erickson et al. 2013; Herfeld 2017). These developments fed into Arrow and Debreu's existence proof of general equilibrium for an economy populated by rational agents (Arrow and Debreu 1954; see Düppe and Weintraub 2014; Kirtchik and Boldyrev 2024). In this setting, "rationality" was distinguished from "reason" or "intelligence" and recast as a formal, axiomatic notion—"rigid rules that determine unique solutions" (Erickson et al. 2013; see also Klaes and Sent 2005). From this axiomatic tradition arose a second meaning of rationality centered on consistent choice: agents are rational insofar as their preferences satisfy formal coherence conditions (Giocoli 2003).

At the level of individuals, the 1940s marked also a decisive turn toward formal and axiomatic models of behavior. In *Theory of Games and Economic Behavior*, John Von Neumann and Oskar Morgenstern (1944) set out axioms on preferences (transitivity, completeness, independence) that support a representation of choices by a value function. This framework underpins expected utility theory and its many variants (Moscati 2023). Leonard Savage extended it to subjective expected utility, addressing decisions under uncertainty when objective probabilities are unknown (Savage 1954).

This modeling invited two readings of "rationality." A positive reading seeks to describe how agents actually behave; a normative reading prescribes coherent decision procedures. As rationality was recast in terms of consistent choice rather than as a psychological portrait of *homo oeconomicus*, the normative strand broadened—from individual behavior to questions of social choice and the design of “rationalizing” policies (Herfeld 2018; Hands 2015).

Another key institution in the history of rationality was the Graduate School of Industrial Administration at Carnegie Institute of Technology. With Office of Naval Research funding in the 1950s, a GSIA team—including Charles Holt, Franco Modigliani, John Muth, and Herbert Simon—was tasked to advance theory of the firm and production–inventory planning (Pedro Garcia Duarte 2009; Klein 2015). This project was pivotal for both Simon’s “bounded rationality” and Muth’s “rational expectations” (Simon 1955; Muth 1961; see Young and Darity Jr. 2001; E.-M. Sent 2002). Initially designed for micro analysis of firm behavior, the rational expectations hypothesis was adopted in the very late 1960s and early 1970s by Robert Lucas, Edward Prescott, and Thomas Sargent (E.-M. Sent 1998; da Silva 2017). The hypothesis quickly started to raise intense methodological and empirical debates—notably in macroeconomics, where it eventually became a foundational component DSGE modeling (Goutsmedt et al. 2019; Sergi 2020)—and in finance (Delcey and Sergi 2023).

Over the same period, expected-utility theory faced mounting challenges from paradoxes associated with Maurice Allais and Daniel Ellsberg, prompting historical and analytical reassessments (Moscati 2024; Zappia 2018, 2021). These critiques fed into Daniel Kahneman and Amos Tversky’s Prospect Theory (Kahneman and Tversky 1979) and the rise of behavioral economics (Heukelom 2014; Truc 2022), while “bounded rationality” attracted renewed attention after the 1980s (E.-M. Sent 2018).

This history of the meanings and uses of “rationality” is also a history of changing methods. It reflects shifts in both mathematical techniques and in what counted as rationality: from a “system-of-forces”—“economic processes generated by market and non-market forces”—to a “system-of-relations,” where the aim is to establish the existence and properties of equilibria via the “validation and mutual consistency of given formal conditions” (Giocoli 2003, 4–5). In this transformation, axiomatization and fixed-point theorem became central (Weintraub 2002; Herfeld 2017). Game theory also gradually gained a place in economics, though it remained marginal in the years immediately after Von Neumann and Morgenstern’s publication of *Theory of Games and Economic Behavior* (Giocoli 2003; Ericsson 2015).

The history of rationality is also one of new alliances with engineering. Work by Holt, Modigliani, Simon, and Muth helped import control theory into economics (Klein 2015), while new classical macroeconomics—Lucas, Prescott, and Sargent—introduced elements of “information engineering” inspired by information theory (Boumans 2020). Finally, this history intersects with the “applied turn” in economics: challenges to expected-utility theory coincided with rising experimental methods that became central to behavioral economics (Backhouse and Cherrier 2017; Heukelom 2014; Truc 2022).

What can quantitative methods add to this rich historiography? Prior work has emphasized pioneering figures such as von Neumann, Koopmans, or Arrow. Yet, as Giocoli (2003, 9) notes, the fact that foundational contributions were made in the 1940s–1950s does not mean that new conceptions of rationality were immediately adopted. Timing and diffusion remain open problems that we address. The publication surge after the 1970s likely multiplied meanings and applications of “rationality,” complicating any synoptic account. Our approach offers a broader view of uses and interpretations beyond canonical texts that explicitly thematize rationality. The challenge is heightened by the concept’s spread across general equilibrium theory, game theory, macroeconomics, behavioral economics, finance, firm behavior, public economics, social choice and welfare, political economy, and institutional economics. Distinct methods, formalisms, and mathematical tools further complicate mapping its evolution.

This article does not seek a definitive history. It offers snapshots showing how quantitative methods can confirm established findings, refine or challenge others, and widen the scope of historical inquiry.

Which methods and which challenges ?

Historians of economics now have a growing repertoire of quantitative tools, including statistical inference, machine learning, textual analysis, network analysis, and bibliometrics. Recent discussions of the "quantitative turn" in the history and philosophy of economics have weighed the merits and limits of these methods, emphasized their complementarity with qualitative approaches, and speculated about their future relationship. Yet much of this debate remains abstract, offering useful overviews but which provide few clues on what quantitative analysis requires in practice. We take a different path by highlighting what kinds of questions and challenges arise at each stage of the research process, grounding our discussion in the study of rationality in economics. This focus lets us highlight (a) the crucial issue of selecting sources and building data; (b) how even simple quantitative assessments can be informative; (c) the challenges and subjective choices involved in building and adapting tools; and (d) why interpreting results demands careful attention to indicators and close knowledge of both the corpus and its historical context. By tracing, step by step, how we assemble and analyze our corpus, we aim to provide a practical example and a reflection on broader methodological issues faced by quantitative historians of economics.

From raw source to data

Discussions on quantitative methods often focus on the varieties of methods existing. In practice, however, analysis and interpretation come last. Most effort goes into collecting, cleaning, and structuring data, much of which remains invisible in published work. Sources rarely arrive as ready-to-use datasets; they must be transformed from raw materials into usable corpora. In short, the quantitative historian must be as much a data wrangler as a data analyst.

The first step is to locate sources. In our case, we seek a comprehensive collection of economic academic documents that may discuss the concept of rationality and we want to know how the concept is actually used in these discussions. Bibliometric databases are a well-structured option: they record relatively well-structured data, systematically categorizing key features of scientific output—such as authors, journals and affiliations, *etc.*. Some of these databases record citation data, which enables the study of intellectual influence, diffusion patterns, and the structure of scholarly communication. A crucial limitation is the absence of full text. Beyond titles—and, for recent periods, abstracts—the actual text where debates occur is usually missing or costly to access. An important exception is JSTOR, which provides freely full texts for roughly 130 economics journals from 1900 to today.

Before going further, we pause on a methodological point: how data availability shapes what we can ask—and answer—about rationality. The scarcity and structure of available data affect every stage of inquiry, from the choice of research questions to the interpretations of quantitative results. In practice, our research may be shaped as much by the availability of our data as by our own research preferences. In our case, Figure 1 shows that our full-text database, like most available databases, disproportionately represents Anglo-Saxon economic journals, which are also those most systematically digitized and preserved. Not only do we tend to overlook other traditions of economic research, but this also raises a serious issue of presentism: the fact that retrospective data on these journals are easily accessible today does not imply that they were equally central in earlier ones, nor that articles were the dominant vehicle for the diffusion of ideas. Rather, it reflects the fact that they had the resources and opportunities to digitize and maintain comprehensive digital archives of their publications. Moreover, several key economics journals are absent from JSTOR (e.g. *Ecological Economics*, *Energy Policy*, *Applied Economics*), while other bibliometric databases such as Web of Science or Scopus present their own limitations in coverage and accessibility.[3]

[3] One of the main challenges for a quantitative history of economics is to move beyond reliance on proprietary

Once access to JSTOR full text is secured, another crucial step is to transform them into a usable database. In most cases, data are not readily available in a structured format; rather, they result from a process that involves cleaning, categorizing, and selectively removing or reformatting information from raw sources in order to produce a dataset suitable for computational analysis. This challenge is particularly true in the history of economics, where the primary sources are the texts themselves—often unstructured and interspersed with various layers of metadata, such as section headings, footnotes, bibliographic references, or editorial annotations.

Data wrangling is not a tedious prelude to "real" historical analysis. Manipulating, cleaning, and structuring raw sources is a fundamental and often generative stage of research. As Lemercier and Zalc (2019, 62) reminds us, collecting and categorizing sources is also "a moment to reflect on the sources and the purpose of the research." Direct engagement with raw data can prompt the reevaluation of initial hypotheses and the emergence of new questions. In our case, preparing full-text documents raises basic design choices about inputs and, by extension, what counts as a relevant economic text. Should we include book reviews, editorials, working papers, or conference proceedings? How should we define an "economics journal"—restrict it to core outlets or extend it to interdisciplinary venues where economics appears regularly? Even within a single article, boundaries are ambiguous: should headings, abstracts, footnotes, appendices, or acknowledgments be analyzed or excluded? Each decision carries historiographical implications. It shapes the corpus and, ultimately, the history of rationality our methods can reveal. There is rarely a single definitive choice, but rather a series of trade-offs that should be made explicit.

Our first choice was to restrict the analysis to English-language articles, since cross-language comparisons involve additional challenges. Indeed, most textual methods are designed to identify commonalities within texts that share a specific linguistic structure. Hence, the same topic discussed by two documents in two different languages will likely not be identified as related because linguistic differences mask underlying semantic similarity. We also restrain our analysis to research articles and filter out book reviews, comments, editorial reports or obituaries.[4] While such materials can illuminate how rationality was debated, they are not primary sites for articulating new ideas within the discipline. Moreover, textual data are by nature voluminous, and filtering improves tractability for large-scale computation. At the document level, we focused on the body text and removed (as far as possible) peripheral elements such as acknowledgements, references, or appendices.

Studying Rationality through Texts

Once the corpus has been assembled, the next challenge is to identify how the concept of rationality is actually used in the texts. A straightforward first step is simply to measure the frequency of occurrence of the terms themselves. Figure 2 shows the relative frequency (with respect to the total number of words published each year) of "rational" and "rationality." The figure reveals a clear surge in the 1980s and 1990s, suggesting that discussions explicitly framed in terms of rationality became particularly prominent during this period.

However, the key question is not whether a document contains the word "rationality," but rather what the term means in specific context. In other words, our aim is to move beyond keyword retrieval and recover the different meanings and intellectual settings in which "rationality" is invoked. A direct way to begin is co-occurrence analysis: examining the words

digital libraries and to actively develop open, historically inclusive corpora. This includes incorporating materials produced in non-Anglophone countries and expanding the range of textual formats considered—such as books, book reviews, working papers, and conference proceedings.

[4] This filtering draws on JSTOR's own classifications, supplemented by our manual checks. Despite these safeguards, a perfectly clean restriction to research articles was not feasible, and some residual non-article items likely remain.

that most often appear immediately before or after "rational" or "rationality." Such co-occurrences indicate the conceptual frames and debates in which the term is embedded. Figure 3 reports, by decade, the five words most frequently adjacent to "rational" and "rationality," showing how associations shift over time. Before the 1940s, the picture was heterogeneous, with links to philosophy, psychology, and general notions of reasoning. From the 1940s onward, the rise of choice theory places rationality at the center of economic modeling as a device for describing and formalizing behavior. By the 1970s—and especially the 1980s—the framework was both extended and contested, with the advent of rational expectations on one side and bounded rationality on the other.

Co-occurrence analysis offers a first view of how talk about rationality changes, but it is limited to the immediate lexical neighborhood of a word and thus to proximity in vocabulary, not meaning. Two sentences may share no terms—"rational behavior" and "profit maximization"—yet point to closely related conceptions of agency and choice. To move from this broad picture to how each document actually mobilizes the concept—and to map semantic relationships not visible through shared keywords—we turn to recent advances in natural language processing: large language models (LLMs). These models encode context and meaning, allowing us to compare sentences that are semantically similar even when their vocabularies diverge, and to see how the same word takes on different meanings across contexts. Quantitative text analysis is not new, but its scope has expanded with these methods. Before introducing our LLM-based approach, we briefly present core principles of text representation, which remain relatively unfamiliar in the history of economics.

One of the central ideas in text analysis is *text representation*: how units of text (words, sentences, documents) are encoded as numbers for computation. The most basic and historically dominant option is the *bag-of-words* representation. Suppose the unit is the document and you want to group similar documents. How can you compare them numerically if texts are words but computers process numbers? Bag-of-words solves this by turning each document into a vector of word counts. Each dimension corresponds to a vocabulary term, and the value is its frequency in the document. Documents that use many of the same words have similar vectors. This is the representation used by many probabilistic topic models increasingly applied in our field (e.g., Ambrosino et al. 2018; Goutsmedt and Truc 2023).

Any text representation transforms a rich source of natural language into a simplified numerical format and, inevitably, loses complexity and nuance. The key question is what it preserves and what it discards. Because bag-of-words keeps only frequencies, it throws away context and word order. A document is represented as a bag in which words are blindly tossed. For example, "Hayek is right and Keynes is wrong" and "Hayek is wrong and Keynes is right" receive identical representations, though their meanings are opposite.

A major advance that mitigates these limits is *word embeddings* (Mikolov et al. 2013). Instead of frequency vectors for documents, embeddings learn a vector for *each* word that captures semantic relations from patterns of co-occurrence in large corpora.[5] Embeddings represent each word as a point in a vector space; words that are closer in the vector space tend to be semantically similar even if they rarely co-occur. Figure 4 illustrates the idea with a two-dimensional toy example: we created artificial word vectors in a two-dimensional space. In practice, embedding spaces are high-dimensional, but the principle is the same: absolute positions are arbitrary, while relative distances encode semantic relationships.

The first generation of embedding models, such as Word2Vec or GloVe, learned static word vectors: each word had a single position in the vector space, regardless of the sentences or documents in which it appears. This makes it difficult to grasp differences in meanings of a same word. The more recent generation of embedding models, known as LLMs, builds on a different principle: rather than assigning a fixed vector to each word, they produce

[5]The intuition behind word embedding is the distributional hypothesis, famously stated by the linguist (1957: 11): "You shall know a word by the company it keeps".

contextualized embeddings—that is, vectors that depend on surrounding words (the context). In LLMs, the meaning of the word "rationality," captured by a word embedding vector, may vary significantly depending on the contexts in which it is used in the studied corpus. In this way, ideas and conceptual transformations become measurable and quantifiable objects, opening new possibilities for the study of economic ideas.

We use Sentence-BERT (Reimers and Gurevych 2019), a variant of the LLM BERT, designed to produce sentence embeddings rather than word embeddings. This model yields a single vector representation per sentence, which allows straightforward semantic comparison by measuring the similarity between sentence vectors. To illustrate how embedding vectors capture semantic information, we compare the real sentences from the database with a synthetic sentence: *"Economic agents are supposed to be rational."* Table 1 reports the most similar sentences retrieved for various years.[6]

We vectorized more than 31 millions of sentences from the JSTOR database using Sentence-BERT. Our objective is to identify sentences that discuss rationality and related concepts. Rather than relying solely on the keyword "rationality," we use cosine similarity between sentence embeddings to detect semantic proximity. For example, if sentence A explicitly mentions rationality and its vector is close to that of sentence B—which does not use the term—then B likely addresses a related idea.

A straightforward approach would have been to compute a single average embedding vector for all sentences containing the words "rationality" or "rational", and to use this as a *representative* embedding for the concept. However, such an average would be present-biased: since the number of documents in the corpus grows sharply over time, the average would be disproportionately influenced by recent decades, thereby over-emphasizing contemporary meanings of *rationality* at the expense of older ones. To account for potential semantic drifts over time, we compute a moving average vector for each year, using a sliding window of five years (i.e., the year in question plus five years before and after). From 1900 to 2020, this yields 120 yearly representative vectors that reflect the evolving contextual meaning of *rationality* in the economic literature. For each year, we then computed the similarity between sentences from the corpus and the respective representative vector and we retain the top 1% most similar sentences (X sentences).

We interpret these sentences as reflecting how economists mobilize the concept of rationality, in different historical contexts. Because of the huge number of sentences we have identified, we would like a method to group together sentences using a similar meaning of rationality. We clustered the sentences using the K-means algorithm—one of the most widely used and straightforward methods for grouping LLM embeddings. Because the distribution of our identified sentences is present-biased (Figure 1), clustering the full set would risk over-weighting recent periods. We therefore run clustering separately for each decade from 1900 onward (merging the first four decades two by two due to small counts). Table 2 reports two clusters—one from the 1980s and one from the 1990s—showing the three sentences nearest each cluster's representative vector. Note that these textual clusters group sentences rather than documents; hence, a single document can belong to several clusters. Despite being arbitrarily separated by decade, it is clear that sentences in both clusters convey a very similar meaning of rationality centered on the concept of rational expectations. To capture this stability of meanings across time, we merge clusters between decades when these clusters are sufficiently close (i.e., in our case, with a cosine similarity superior to 0.7). Following this procedure, the two clusters presented in Table 2 are ultimately merged into a single,

[6] One issue with using LLMs is that they are pre-trained on recent data, which raises the methodological problem of *presentism*. Contextualized embeddings partially mitigate this issue, as the vectorization of texts depends on the immediate linguistic context. As illustrated in Table 1, each historical period is associated with its own specific set of questions and, therefore, with particular contextual usages of the term *rationality*. However, the way these surrounding words influence the vector representation of *rationality* remains shaped by a bias toward present-day language and meanings. This limitation is discussed further in the conclusion.

temporally continuous cluster. Figure 6 displays the complete set of results in the form of an alluvial plot.

To ensure the stability of our results, we compare them with a bibliometric analysis of the corpus. Such a comparison is not costless, but it is highly recommended for at least two reasons. First, crossing the results of advanced computational methods such as LLM embeddings serves as a form of robustness check.[7] Second, the systematic comparison of sources and methods is a fundamental principle of historical analysis, needed to triangulate information. For instance, we saw that our JSTOR full-text database, though large, does not index several crucial recent journals in economics—a gap that can be partly compensated by crossing it with citation databases. Moreover, while full texts provide invaluable insights into the meanings and contexts of concepts, citation data highlight patterns of direct intellectual influence and diffusion. On the one hand, we focus on the semantic uses of rationality and related discussions in the text themselves, independently of authors' institutional proximity or of their citation practices. On the other hand, with bibliometric analysis, we look rather at how these ideas circulated and have been diffused across fields.

Studying rationality through citations

Beyond textual analysis, another important source of information is documents metadata (e.g., authors, journals, institutions, citations). We focus here on citations to study how articles on rationality cite and are cited. Citations trace ideas' lineage and diffusion. The evolution of an idea often depends as much on how it is circulated and appropriated by readers as on how it was formulated by its authors.

However, one major issue for quantitative analysts using this method is temporal: systematic citation practices are relatively recent, so citations present poor quality and reliability before the 1960s. In contrast to textual analysis, thus, bibliometric analysis in economics is largely confined to the postwar period. A second constraint is data scarcity and quality: extracting and standardizing references at scale is complex and imperfect. For this reason we rely on Web of Science for structured citation data, despite its proprietary cost and access limits. We then augment our JSTOR full-text corpus with Web of Science citations and with citation and abstract data for key journals missing from JSTOR.

Citation data does not require sophisticated analysis to be relevant, and can be easily incorporated into the everyday toolset of historians of economics. Simply counting citations can be used to proxy engagement and better understand the impact of a particular author or publication. There are clear reasons to incorporate citation studies into historical research. Beyond tracking diffusion, highly cited papers are more visible and attract more engagement (Matthew effect). Recent work also suggests that citations shape reading behavior and perceived quality: highly cited papers are more likely to be read closely and to be seen as substantial intellectual influences (Teplitskiy et al. 2022).

Historians routinely discuss scientific influence, and recognition using proxies such as major grants, prizes, and honors. For example, E.-M. Sent (2004) narrative of the transition from the dominance of rational choice, through the limited success of "old" behavioral economics (e.g., Simon, George Katona), to the rise of "new" behavioral economics is organized around such milestones. In this sense, citations serve as a complementary proxy alongside traditional markers. In a large qualitative–quantitative study of the Nobel Prize, Offer and Soderberg (2016) distinguished several profiles: laureates who peak at the prize then decline, "innovators with staying power," "still rising" winners honored before their citation peak,

[7] Sophisticated techniques used in the history of economics—whether topic modeling, network analysis, or contextual embeddings—often depend on modeling choices (for example, the number of topics in a topic model) that can substantially affect the output. Since these outputs are never self-interpreting and always require human qualitative assessment, it is time consuming to conduct robustness checks.

and late winners recognized long after their peak. Relating institutional rewards to citation and publication patterns clarifies how recognition interacts with diffusion, reception, and appropriation.

Figure 7 plots citation patterns for four seminal references on bounded rationality across all economics journals and the top five. The selection is partly arbitrary but standard in the literature: Simon (1955) and Allais (1953) are early critiques of neoclassical rational choice with both empirical and normative implications, while Akerlof (1970) and Kahneman and Tversky (1979) are early "new" behavioral landmarks that mark the field's emergence and growth.

The influence of "new" behavioral economics overwhelms that of "old" behavioral economics, in both the top five and the full set of economics publications indexed in Web of Science. Whereas Simon (1955) and Allais (1953) are never cited by more than 0.20% of all economics-article, Kahneman and Tversky (1979) reached at least 1% by the late 2010s and continues to rise. The contrast is both in scale of influence and speed of acceptance: citations to the two "new" behavioral papers grow rapidly and steadily right after publication, whereas Simon and Allais peak around the time of their Nobels or only much later in the 2010s wit the emergence of "new" behavioral economics. As argued by Offer and Soderberg (2016), many laureates receive a "Nobel premium," a modest post-prize citation bump. Simon and Allais fit this pattern, but for Kahneman and Tversky the effect is extreme: after the Nobel, their declining trend reverses and climbs throughout the sample. (Offer and Soderberg 2016)

Citation counts are a blunt tool and can be extended in several ways. One can examine *who* cites a work—by discipline, journal, or subfield—and assess the *qualities* of citations, distinguishing positive from negative uses or functional roles in the text (Budi and Yaniasih 2023). Another way to use citations is to transform citation data into relational data. A prominent example is bibliographic coupling, widely used in the history of economics. It maps relationship between articles by using the similarity of reference lists: articles are nodes, and the more references two article share, the closer they appear in a two-dimensional layout. The premise is that shared references proxy intellectual proximity. These maps help delineate the frontiers of disciplines, fields or sub-specialities and expose their internal organization, including hierarchical and/or core-periphery structures.

Running bibliographic coupling begins with delimiting the corpus: which articles count as "about rationality"? Keyword searches are brittle, since papers on revealed preference or prospect theory may not use the words "rational" or "rationality." Instead, we use our LLM sentence embeddings from JSTOR. For each article in the coupled JSTOR–Web of Science database, we compute the cosine similarity between its article embedding (the average of its sentence embeddings)—or, for missing journals in JSTOR, on the abstract embedding—and the year-specific representative vector. We retain the closest 10% of documents in this space, selecting those most engaged with uses of rationality in that year. Because embeddings encode meaning rather than only shared vocabulary, this procedure captures papers that participate in rationality debates even when the term is absent. It yields a bibliometric corpus that goes beyond keyword search and traces the wider intellectual conversation around rationality.

We apply a dynamic network analysis (Goutsmedt and Truc 2023), which parallels our sentence-clustering method. Rather than a single network for the entire period, we build overlapping networks for shorter windows to preserve historical resolution. A clustering algorithm is run on each window, then communities are merged across windows when they share enough nodes (documents). In bibliographic coupling, a community is a set of documents that share a substantial fraction of references, indicating intellectual proximity. Our bibliometric communities thus group documents that are likely to engage with rationality in similar ways. By contrast, our textual clusters capture semantic similarity at the sentence level.

After these steps, a central question remains: how do close readings and qualitative analysis of primary texts shape interpretation?

Do you still need to read ?

A common criticism is that quantitative analysis distances researchers from economic ideas. This view is misleading for several reasons. First, as discussed above, data wrangling requires close engagement with texts. Turning raw sources into structured data involves consequential choices about cleaning, categorizing, and exclusion, which demand contextual knowledge of the corpus and its production.

Second, quantitative methods do not yield objective, ready-made outputs. In our case, they produce statistical groupings—bibliometric communities and textual clusters—that require qualitative interpretation. This is a classic strategy to reduce a large and complex set of texts to a manageable number of simple categories. But these clusters are constructed on the basis of statistical commonalities, whose actual conceptual meaning can only be characterized through qualitative assessment. More importantly, a good qualitative knowledge of the corpus and of intellectual debates at stake are indispensable to make sense of raw results. Thus, whether in the collection of data or their analysis, quantitative methods require a constant back-and-forth with qualitative reasoning.

This is especially important because our methods are "unsupervised." With no prior labels to guide classification, human interpretation is central to assessing validity and usefulness. Careful validation—via quantitative indicators and qualitative checks—is essential. The practical question is whether our categories form a helpful classification. A key issue here concerns the construction of indicators that help analysts decide which documents should be read as a priority. Indicators will depend on the corpus and the research context, but a sound strategy is to rely on a combination of indicators rather than on a single measure. It is also crucial to design indicators using information available from the historical period under investigation in order to avoid presentism. For example, using total citation counts to understand a past community can mislead the scholar, since such counts reflect current popularity rather than contemporaneous influence. Equally important, such indicators should be openly shared with other scholars to ensure transparency and replicability of the analysis. To put these principles into practice, we built an open-source dashboard—available here: https://0199205a-7b17-ab11-595d-8f12aae160dd.share.connect.posit.cloud—which compiles the indicators we relied on to interpret clusters and communities.

In the case of a large-scale study such as this one, most results are not surprising—and there is nothing wrong with that. On the contrary, given the large body of works that have contributed to the historiography of rationality, one should expect our method to confirm existing findings rather than to produce entirely new discoveries. Precisely because much of what can be uncovered through quantitative analysis has already, in some way, been identified by other means, these methods can be trusted. This confirmation provides the basis for using them to explore where the historiography is less developed.

Discussion

How can these complex quantitative methods and their careful interpretation contribute to our understanding of the history of rationality? How can they help enrich, complete, and refine our understanding of the various and evolving meanings of rationality in economics? This paper does not propose an alternative history of the concept, nor a comprehensive account of its evolution. Its aim is to show, concretely, the benefits, limits, and uses of quantitative methods in the history of economics.

We highlight selected findings that confirm and strengthen strands of the existing literature, broaden its scope, and open paths for further research. We combine multiple indicators with qualitative analysis, reading articles the models pointed out. Our goal is to show how we navigate results and triangulate indicators to produce clear insights and coherent narratives.

Retrieving known patterns from the history of rationality

Several patterns in our results confirm established consensus in the historiography. First, as rationality theory developed, economics moved away from psychology (Giocoli 2003). Analyzing figures such as Vilfredo Pareto, Irving Fisher, Lionel Robbins, and Paul Samuelson, Giocoli reconstructs the long-standing debate over the relation between economics and psychology. This debate about the role of hedonism in explaining behavior appears clearly in Cluster 2 (Figure 6), which extends from 1900 to the 1940s. The cluster brings together advocates of hedonistic foundations and their critics, notably institutionalists who questioned psychological grounding. Wesley C. Mitchell (1916, 160) illustrates this opposition in one of the closest sentences to our representative vector: “to find the basis of economic rationality in the development of a social institution directs our attention away from that dark subjective realm, where so many economists have groped, to an objective realm, where behavior can be studied in the light of the common day” .

The 1940s and 1950s reveal two textual clusters—highly correlated but below our merge threshold—centered on debates over economic freedom and free enterprise versus economic planning. These clusters include roundtables and special issues on planning, such as a discussion sparked by Oskar Lange’s 1949 article (see Perroux et al. 1949), with contributions by François Perroux, Jan Tinbergen, Evsey Domar, and Michał Kalecki. They also include debates on “the proper spheres of individual freedom and collective control in the ‘good’ economy” (Taylor 1948), in a collection that included, among others, Henry Simons. Across both clusters, Frank Knight appears as a recurrent reference.[8] Somewhat linked to this is the textual cluster 53, which from the 1950s on discussed social choice and welfare, with rationality framed as a guide for decision makers.[9]

The postwar period also hosts intense debate on utility theory—cluster 31 in the 1940s, notably Friedman and Savage (Friedman and Savage 1948), and especially after 1950 in cluster 46. This work revisits demand theory, questions the use of cardinal utility, and scrutinizes the possibility of interpersonal comparisons. Chicago economists such as Friedman and Savage (Friedman and Savage 1952), Stigler (Stigler 1950), and later Becker (Becker 1962) are prominent. In parallel, Nicholas Georgescu-Roegen offered a sharp critique of “utility” and its measurability (Georgescu-Roegen 1954).

For the most recent period, a broad overview of our results shows two expected patterns. In both bibliometric and textual analyses, from the 1970s rational expectations became central and occupied substantial discussion, appearing across multiple textual clusters and bibliometric communities (Figure 6; Figure 8). From the 1980s onward, challenges to the standard rationality approach embodied by expected utility theory emerged in several textual clusters (e.g., clusters 83 and 93) and persisted until the end of our period. A distinct behavioral economics cluster (106) appeared in the 2000s. In the twenty-first century, these themes became more prevalent than rational expectations. Similarly, bibliometric analysis shows a first community labeled “Behavioral Economics and Choice Theory” in 1977–1984, followed by many communities after 2000 that form a dense network around the issue of how to model rationality (Figure 8).[10]

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[8] See also textual cluster 31 on planning issues.

[9] Kenneth Arrow, Herbert Simon or James Buchanan are central in these debates.

[10] See for instance the communities “Behavioral Economics and Experiments”, “Ambiguity and Uncertainty”, “Intertemporal Choice and Control” or “State Preference Valuation”.

A rational expectations revolution ?

One focused read of our results centers on "rational expectations." The concept has an early history: developed by Muth (1961) for price movements in agriculture; popularized when Carnegie colleagues—Robert E. Lucas, Edward C. Prescott, and Thomas J. Sargent—applied it to macroeconomics (Robert E. Lucas Jr. 1972; Thomas J. Sargent and Wallace 1973). It became central in the early 1970s and quickly controversial for monetary and fiscal policy (see e.g., Thomas J. Sargent and Wallace 1975). By the late 1970s it circulated in policy and the press and was often described as a "rational expectations revolution" (Pedro G. Duarte 2025).

Our methods help account for the rapid diffusion beyond controversial policy debates during the stagflation era. First, the results show no significant trace of rational expectations in the 1960s in either bibliometric or textual outputs.[11] From the 1970s it has taken a central place in the literature on rationality.

The controversial character of rational expectations appears first in the text analysis. Two clusters on rational expectations emerged in the 1970s (textual clusters 67 and 72). The first gathers articles that debate the hypothesis itself, including many critiques of its theoretical and empirical relevance in the 1970s–1980s. Others examine the implications of adopting the hypothesis across domains. The second focuses on models that use rational expectations rather than the hypothesis per se.[12] Here, alongside promotion of such models—Sargent is a major contributor—there are early critiques targeting both the rational-expectations assumption and additional auxiliary assumptions. A clear example is Ray Fair's complaint about "one class of macroeconomic models that have recently been developed" which rely on "(1) the assumption that expectations are rational, given the available information; (2) the assumption that information is imperfect regarding the current state of the economy; and (3) the postulation of an aggregate supply equation in which aggregate supply is a function of exogenous terms plus the difference between the actual and expected price level" (Fair 1978, 411).[13] Diffusion is also visible in clusters that predate the use of rational expectations but incorporate it from the 1980s onward, notably cluster 76 on monetary economics and cluster 28 on price theory.[14]

Our bibliometric analysis complements the text analysis. By tracking references and using overlapping windows, it detects the emergence of new communities and the split of others with fine temporal resolution. The main rational-expectations community, "Rational Expectations and Business Cycles", first appeared in 1966–1973. It focuses less on the hypothesis itself and more on its macroeconomic policy implications. The controversy centers on modelling the inflation–unemployment trade-off and the implied (in-)effectiveness of monetary policy. This aligns with the early divide between "old" and (future) "new" Keynesians (Goutsmedt et al. 2019).

The controversial status of rational expectations in the 1970s–1980s likely helps explain the prominence of this community in our results. These controversies are well documented (Hoover 1988; Goutsmedt et al. 2019). At the same time, our evidence suggests diffusion by

[11] This highlights a key caveat. Because these methods foreground statistically "significant" patterns and indicators' prevalence, they are not always well suited to tracing and interpreting the origins of a concept or theory with the care that close historical study and archival research provide.

[12] The second textual cluster is less centered on rational expectations than the first. It aggregates model-focused debates—some on rational behavior in general, others on demand—within which rational-expectations models remain prominent.

[13] The mix of defense and opposition continues into the 2000s–2010s: we see discussions of DSGE models , alongside macro models with bounded rationality or learning and arguments for behavioral macroeconomics and rational inattention.

[14] Other clusters linked to rational expectations cover inflation expectations and the term structure in the 1970s (cluster 78), forecasting (81), wages and unemployment (82), finance (87), information (89), econometric issues (92), and a more theoretical intersection of rational expectations and game theory from the 1980s (88).

application across domains. The bibliometric analysis indicates two broad pathways. First, communities initially outside the "Rational Expectations and Business Cycles" community began to incorporate the assumption. Second, new communities gradually branched off from the core community: articles that first co-located with the main group increasingly coalesced into autonomous clusters.

Focusing on the first pathway, one community engaged with rational expectations slightly earlier than the "Rational Expectations and Business Cycles". This community addressed topics in trade, demand, and investment within a general-equilibrium framework. Muth (1961) belongs to this community and remains a key reference, even though the rational-expectations hypothesis is not central. In 1971–1978, the label shifts to "Investment and Economic Growth," where Lucas's early work on investment is influential, albeit without using rational expectations. Several important references in this window do employ the assumption, including Cyert and DeGroot (1974) and Townsend (1978). In 1975–1982, the community transformed into a new community, "Investment and Uncertainty," now featuring Kydland and Prescott (1977), Kydland and Prescott (1982), and Robert E. Lucas (1978).[15] Thus, beyond the well-known debates on business cycles and inflation, parallel communities—only partly focused on macroeconomic issues—were active in the early 1970s.

After the mid-1970s, several independent communities began to adopt rational expectations. A first trajectory appears in 1976–1983: two finance-oriented communities ("Asset Pricing and Consumption" and "cl_319") partially merged into "Rational Expectations and Market Information." At its core was imperfect information, with Rothschild and Stiglitz (1976) as a key node. Another central contribution was Grossman and Stiglitz (1980), which drew on Lucas's imperfect-information framework (Robert E. Lucas Jr. 1972) and combined rational expectations with noisy signals to question market efficiency (see Delcey and Sergi 2023). A second community emerged in 1979–1986 with "Game Theory and Information," where rational expectations became more prominent. Here Kydland and Prescott (1977) and Barro and Gordon (1983) on time inconsistency sit alongside Kreps and Wilson (1982) on sequential equilibria and Selten (1975) on equilibrium refinements. In the early 1980s this community split, yielding "Monetary Policy and Inflation," which leverages game-theoretic models to deal with credibility and reputation in policy design.

A second diffusion process involves the emergence of new communities organized around rational expectations that gradually separate from the initial core. In the early 1970s, an international macroeconomics community branched off from "Rational Expectations and Business Cycles." Although it addressed a range of international macroeconomic topics, the determination of exchange rate dynamics—and the role of the rational-expectations hypothesis in that determination—quickly became central, notably with Dornbusch's overshooting model (Dornbusch 1976). From 1974–1981, another community grew out of the core—"Inflation, Expectations and Interest Rates"—which concentrated on the term structure and the use of interest rates to forecast inflation. This community became a meeting ground for macroeconomics and finance, where rational expectations and efficient markets were closely linked (Delcey and Sergi 2023).

This issue of the diffusion of rational expectations merits a dedicated study. Here, the goal is to illustrate the diversity of trajectories and the fine-grained mapping our analyses produce, enabling productive interaction between quantitative evidence and qualitative interpretation.

By the early 1980s, roughly half of the network consisted of communities engaging, to varying degrees, with rational expectations. At the same time, a behavioral economics community had emerged, though it still formed a small share of the network, a situation that changed after the 1990s.

[15] These articles received a good number of citations from the paper of this community, but they appear as "connector" in the sense that they were highly connected to nodes in other communities.

The Trajectory of Simon's contributions and the birth of behavioural economics

It is well known that Herbert A. Simon's "bounded rationality" is one of the most influential concepts related to rationality and underpins a significant critique of the standard treatment of rationality in economics. Its broad uptake, however, was delayed well beyond Simon's initial articles (Simon 1955, 1959). Our bibliometric and textual analyses trace this trajectory in detail.

Simon is a prominent presence in the 1950s debates on rationality. His two seminal articles and his Nobel lecture (Simon 1979) appear across numerous textual clusters (including social choice, the theory of the firm, and price setting), indicating how his analysis of economic rationality resonated across a wide spectrum of discussions. Yet despite this visibility—and despite the Nobel recognition by the late 1970s—his concepts of "bounded rationality," "procedural rationality," and "satisficing" remained marginal at the time. Figure 7 shows the delayed reception: citations rose after the 1970s and reached a substantial share only from the 1980s onward.

Textual analysis provides further evidence. Textual cluster 40, one of the largest in our study, spans roughly seventy years from 1950 onward and aggregates general discussions on rationality in economics. In the 1950s, while Simon (1955) recurs within the cluster, debate centers on rationality in game theory—for example, Schelling (1959) and Ellsberg (1956)—and on expected-utility theory, with contributions by Marschak (1950) and Chernoff (1954). By the 1970s, "bounded rationality" becomes the cluster's most prevalent expression alongside terms such as "behavioral theory". In the 1990s, articles discussed "boundedly rational" agents, "unbounded rationality," and "procedural rationality."[16] Simon's concepts are clearly central to this cluster by then.

The bibliometric analysis, using successive time windows, complements this picture by tracing the destiny of small communities. In the first window (1960–1967), Simon (1959) is cited within a community centered on operations research (cl_5). This community, which persists—though smaller—until 1969–1976, is gradually populated by organizational-theory work critical of the profit-maximization hypothesis in neoclassical models. Key examples include Winter (1971), which applies satisficing to firm behavior, and Cyert and George (1969). A related strand is Leibenstein (1966) on "X-efficiency," which challenges economics' focus on allocative efficiency. Together, Simon, Winter, Cyert (with George), and Leibenstein form a diverse critique of how rationality—especially through profit maximization—is employed in economics, from an organizational theory perspective.

After 1969–1976, this stream landed in a smaller, short-lived community (cl_175). References to Simon and critiques of profit maximization then remain scattered, moving through several small communities (cl_182, cl_251, cl_300, cl_328). This instability underscores the marginal position of this literature and its brief association with other lines of work. In the final community of this trajectory (cl_328), organizational critiques intersect with the theory of the firm in transaction costs and property rights: Williamson (1979) figures here, and Coase (1960) is an important reference. These strands then coalesce into a large 1977–1984 community, "Behavioral Economics and Choice Theory," where Simon appears alongside (Kahneman and Tversky 1979), bringing together multiple critiques of rationality in economics.

From 1979–1986, a more specific "Behavioral Decision Theory" community forms around Kahneman and Tversky's contributions. It persists through the end of our period and, as with rational expectations, spins off several autonomous communities. In addition to the main Kahneman and Tversky's cluster, we find four other clusters about pro-social behavior, behavioral intertemporal decision, behavioral game theory, and behavioral finance. After the

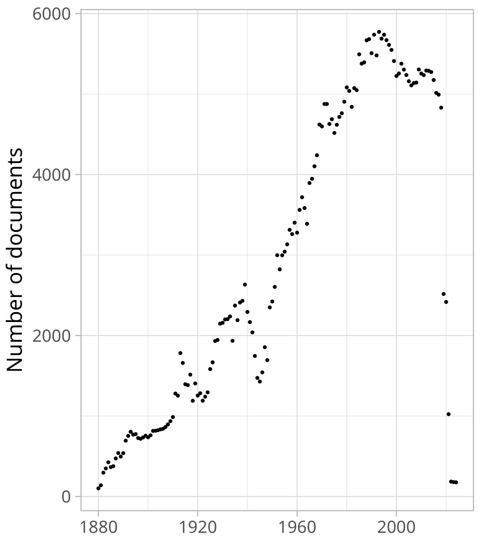
[16] Simon himself became one of the most prevalent words of the cluster.

2000s, behavioral economics becomes a dominant venue for research on rationality with five behavioral economics clusters culminating to around 30% of the network. By contrast, Simon's articles fades as a bibliometric anchor: "bounded" and "procedural" rationality are regularly invoked, as the textual analysis highlighted, but no later community clearly bears his heritage.

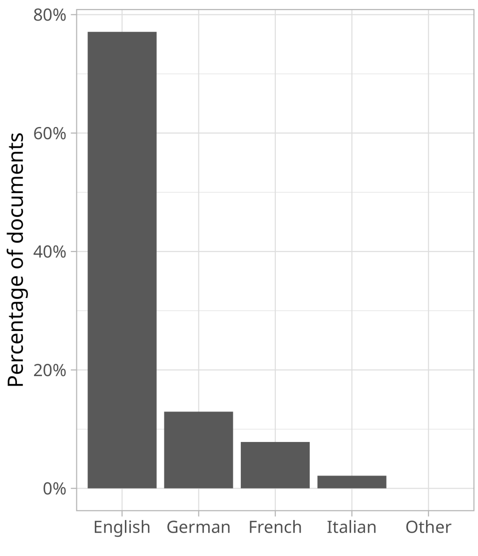
Tt is often argued that the rise of "new" behavioral economics overshadowed earlier "old" behavioral works, and our quantitative studies delineates the specific chronology and magnitude of the different attempts at stirring a behavioral shift in economics. However, as made clear by Figure 7, it is not just a linear story of a shift from one program to the other. Citations to Simon (1955) have never been higher than since the ascent of the new behavioral economics. This resurgence admits at least three interpretations. A more favorable reading is that the "new" program helped revive abandoned research directions and moved toward reconciliation with earlier strands (something promoted by E.-M. Sent (2008) or more recently by Earl (2022)). A more unfavorable view sees intellectual appropriation, in which classic references are reframed to fit the new agenda, renewing interest but with a biased and presentist lens (Mongin 2019). A third more critical interpretation is that the citation increase reflects a growing backlash against the "new" program from the standpoint of "old" behavioral economics. Preliminary evidence supports the second interpretation. This is suggested by the sparse and unstructured citations patterns to "old" behavioral economics and the fact that discussions of rationality are increasingly framed by "new" behavioral and experimental concepts. While a definitive conclusion is beyond the scope of this paper, our study provides a roadmap for future quantitative and qualitative research needed to further the understanding of the relationship between "old" and "new" behavioral economics.

Conclusion

Figures



(a) Documents



(b) Languages

Figure 1.: General statistics of the JSTOR full-text database

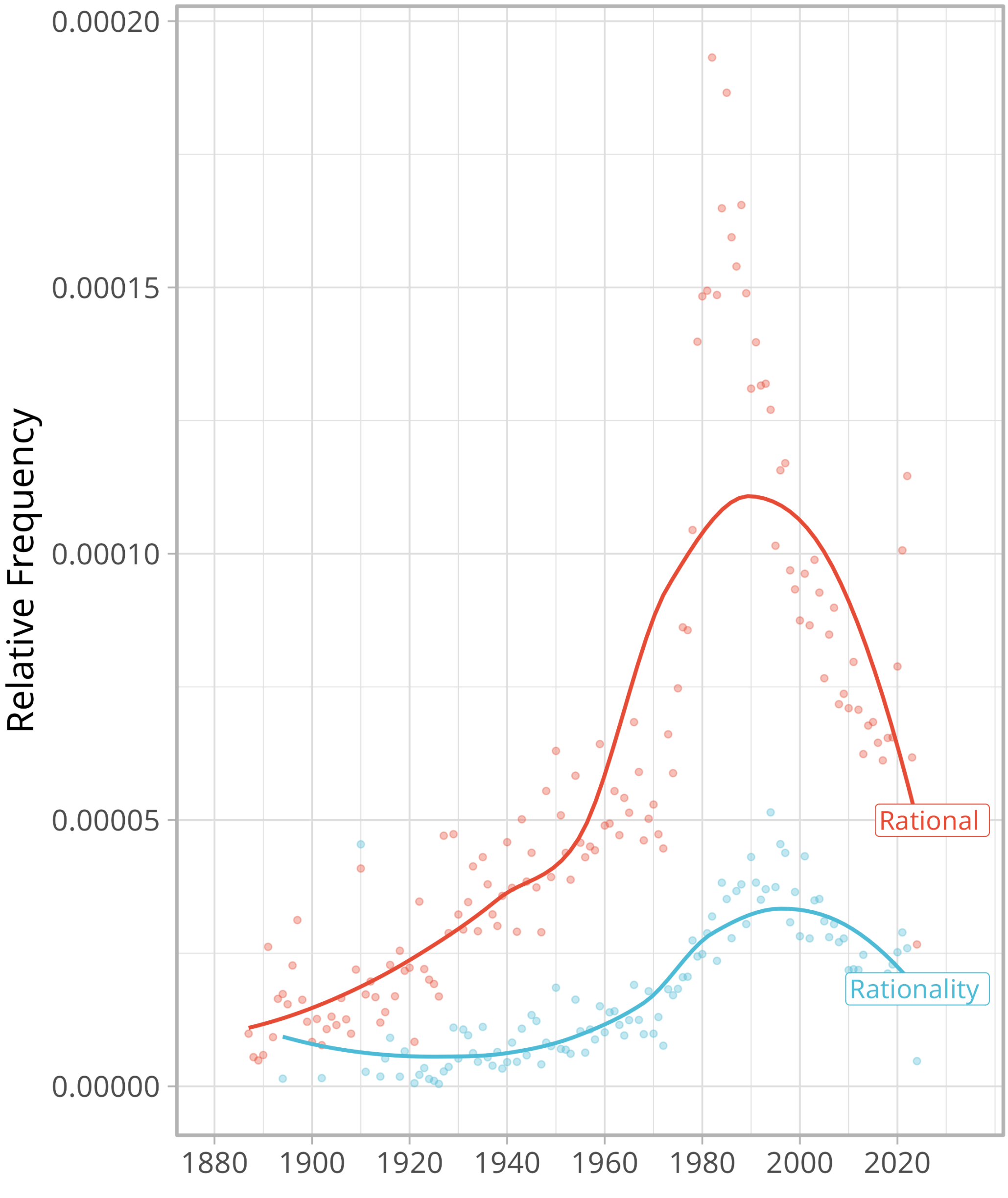


Figure 2.: Relative frequency of terms “rational” and “rationality” in the jstor database

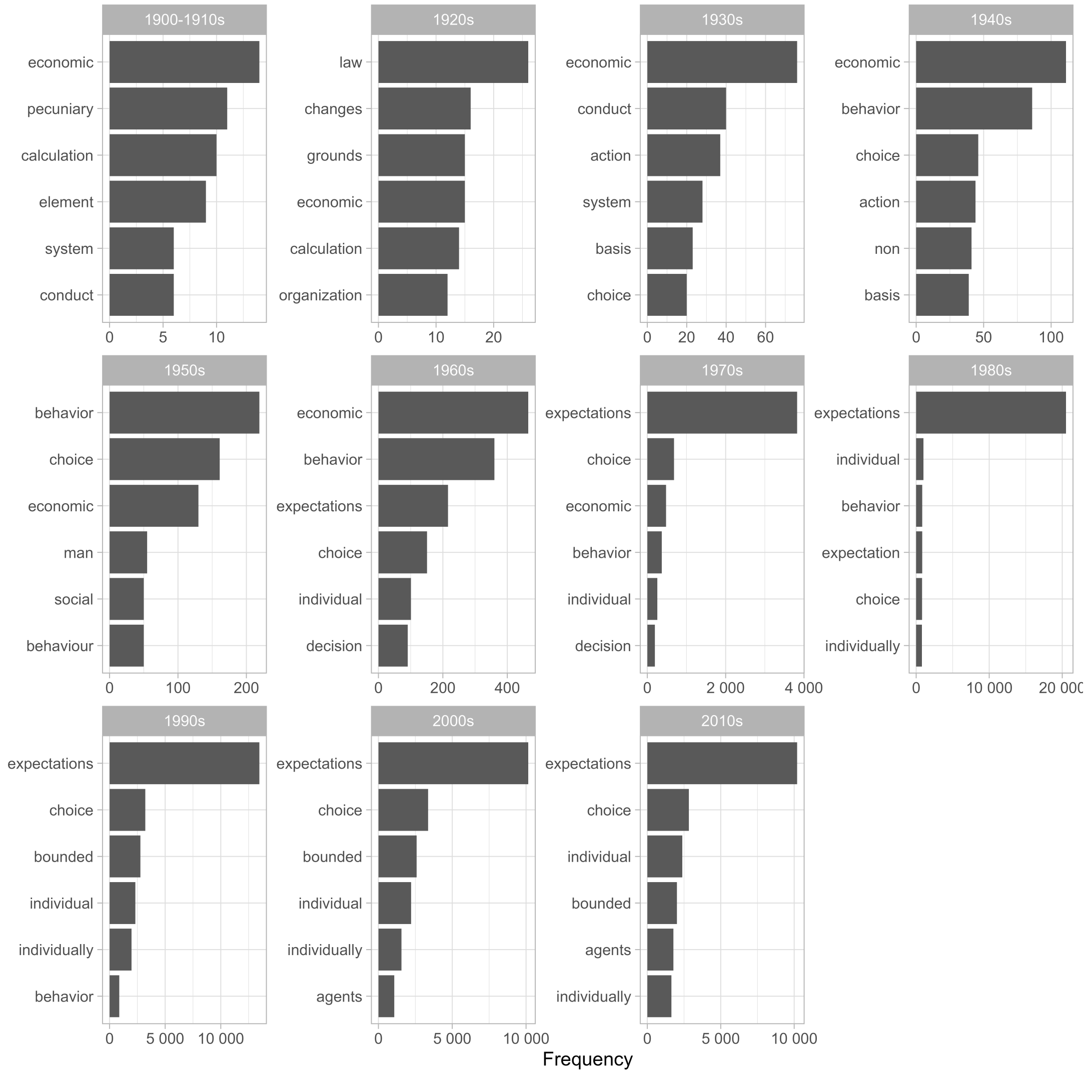


Figure 3.: Most frequent neighbor terms of “rational” and “rationality” by decade

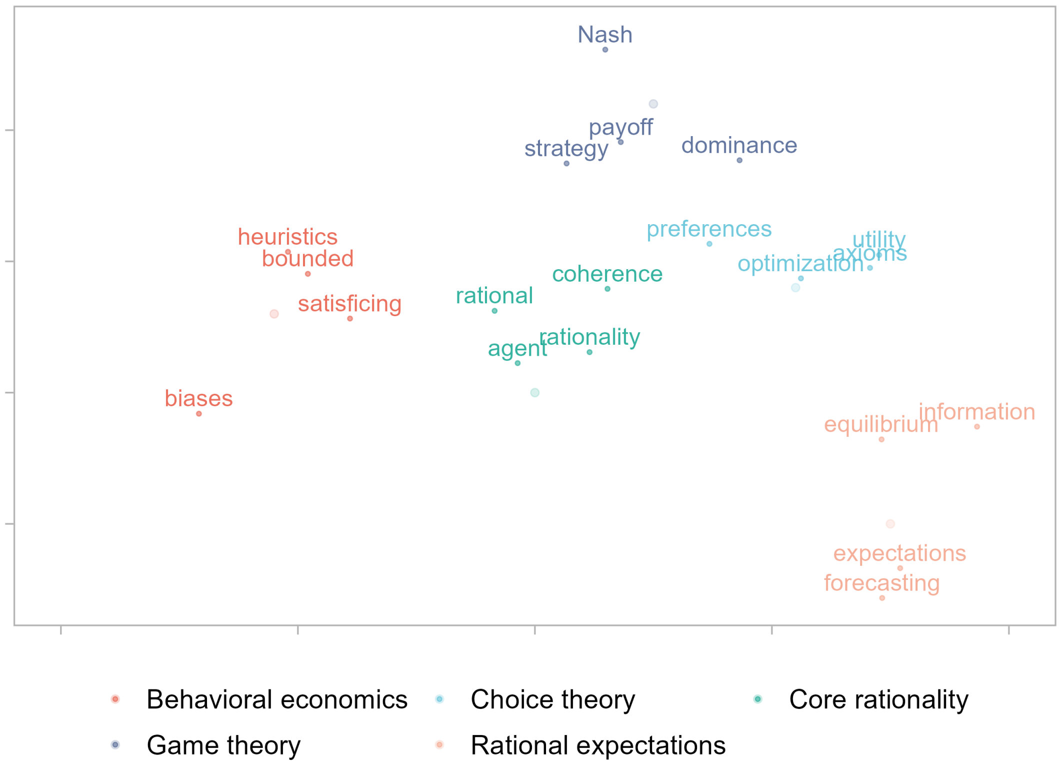


Figure 4.: Illustration of word embeddings in two dimension space

Table 1.: The three closest sentences to the synthetic sentence for selected years.

Year	Sentence	Cosine
1925	8 "The Rationality of Economic Activity," Journal of Political Economy, vol.	0.718
1925	On the other hand, there are important and necessary elements in economics which are not to be subsumed under "human behavior."	0.637
1925	Now no man can possibly give an account of economic behavior without having some working motives of human nature on the back of his tongue.	0.617
1950	;2 "Basically, for Professor Hicks, as for Jevons and Marshall, economic decisions are made by an abstract calculating machine";3 "I find myself continuing to believe ... that, whatever the origin of the want, when it rises to consciousness, it is subject to a process which we call rational, when the human agent decides .	0.706
1950	Modern civilization is lopsided in its emphasis on rationality, especially on economic rationality.	0.700
1950	Rational behavior means conduct motivated by the goal of maximization of money income, wealth and economic advantages.	0.700
1975	Rationality and Irrationality in Economics.	0.750
1975	Such agents are assumed to act rationally and exclusively on the basis of market given information.	0.721
1975	In Marshallian analysis, economic agents are conceived to be not so much rational as reasonable.	0.710
2000	Agents have rational expectations.	0.771
2000	Rational in the narrow economic sense of profit maximizing.	0.768
2000	Actors who operate in an economy are generally rational in the sense that they are interested in improving their individual position.	0.768
2015	Introduction For over one hundred years, economists have operated under the assumption that agents are rational.	0.836
2015	It is far from clear that the same is true if we replace fully rational with behavioral agents.	0.752
2015	On the other hand, an agent might form rational expectations.	0.750

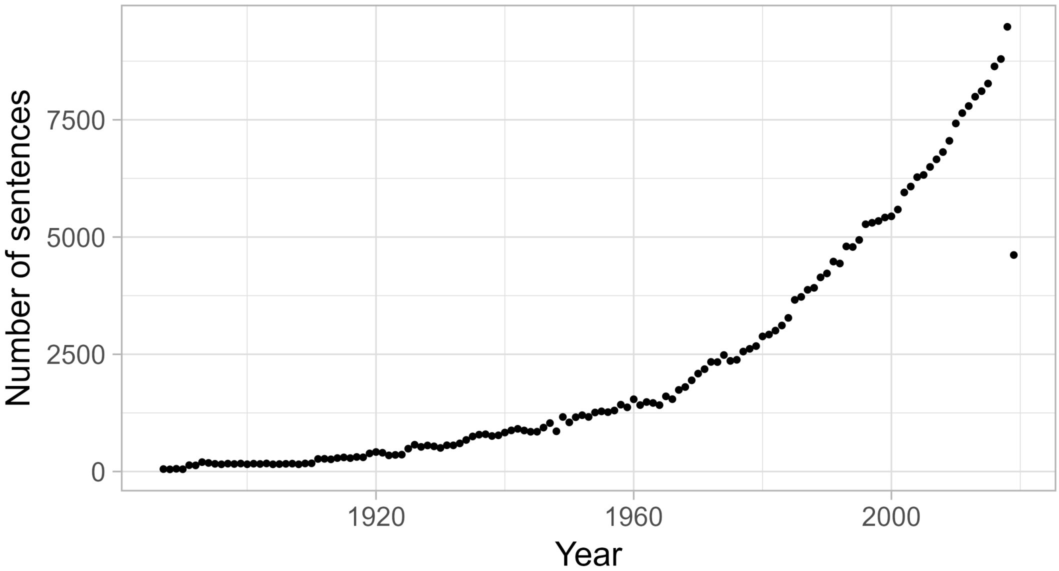


Figure 5.: Distribution of cosine similarities to the fake sentence

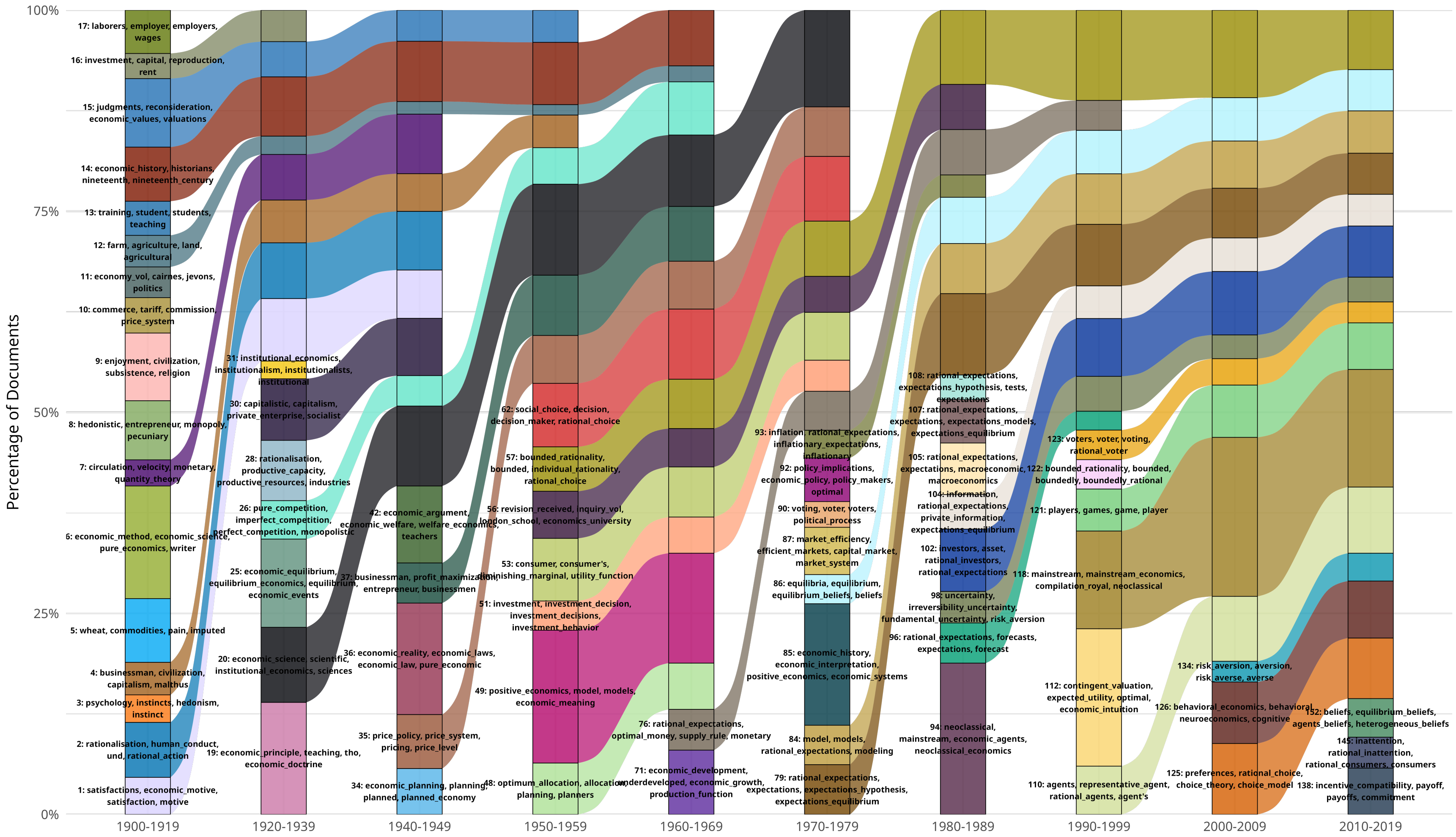


Figure 6.: Alluvial plot of clusters of sentences about rationality

Table 2.: The three closest sentences from two selected clusters (1980s and 1990s).

Sentence	Similarity	Title	Year	Journal	Authors
1980s					
INTRODUCTION The burgeoning literature employing or alluding to the concept of rational expectations in time-dependent economic processes is strangely vague regarding possible mechanisms by which the economic agents could actually achieve the hypothesized ”rationality.”	0.856	Rational Expectations and Learning from Experience	1979	The Quarterly Journal of Economics	Stephen J. DeCanio
Perhaps the justification for this paper is that both critics and some practitioners of the rational expectations approach seem unaware of these implications.	0.838	Rational Expectations and Economic Thought	1979	Journal of Economic Literature	Brian Kantor
A rational agent draws on an information set larger than historical prices, and expectations are rational if they depend upon the same things that economic theory suggests determines the variable.	0.837	A Semi-Strong Form Evaluation of the Efficiency of the Hog Futures Market	1979	American Journal of Agricultural Economics	Raymond M. Leuthold, Peter A. Hartmann
1990s					
This article will criticize rational expectations models for several different reasons.	0.877	What Is so Natural about the Natural Rate of Unemployment?	1981	Journal of Economic Issues	Robert Cherry
In this article we concentrate on the role of the rational expectations hypothesis in the debate.	0.874	THE CONTROVERSY OVER RATIONAL EXPECTATIONS	1981	National Institute Economic Review	David G. Mayes
For the purposes of this discussion, it will be assumed that economic agents form rational expectations.	0.873	The Market for Innovations and Short-Run Technological Change: Evidence from Egypt	1988	Economic Development and Cultural Change	John M. Antle, Charles C. Crissman

Note:
Cosine similarities from the representative vectors; three nearest sentences per selected cluster/decade.

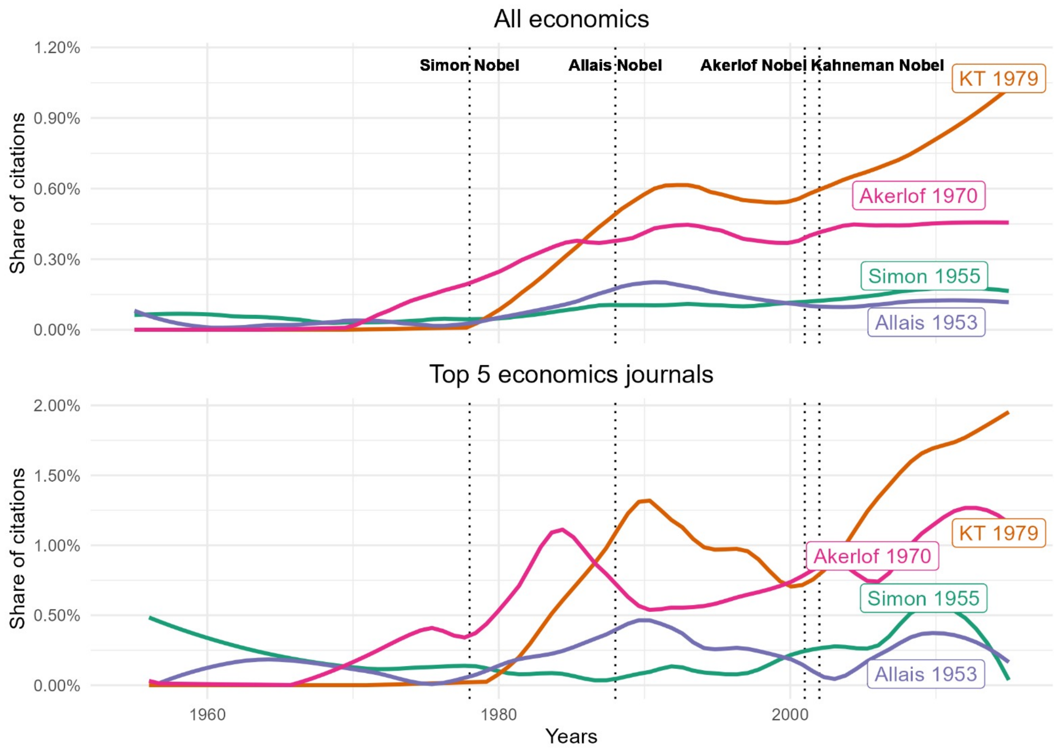


Figure 7.: Relative citation of a selection of seminal papers on behavioral economics

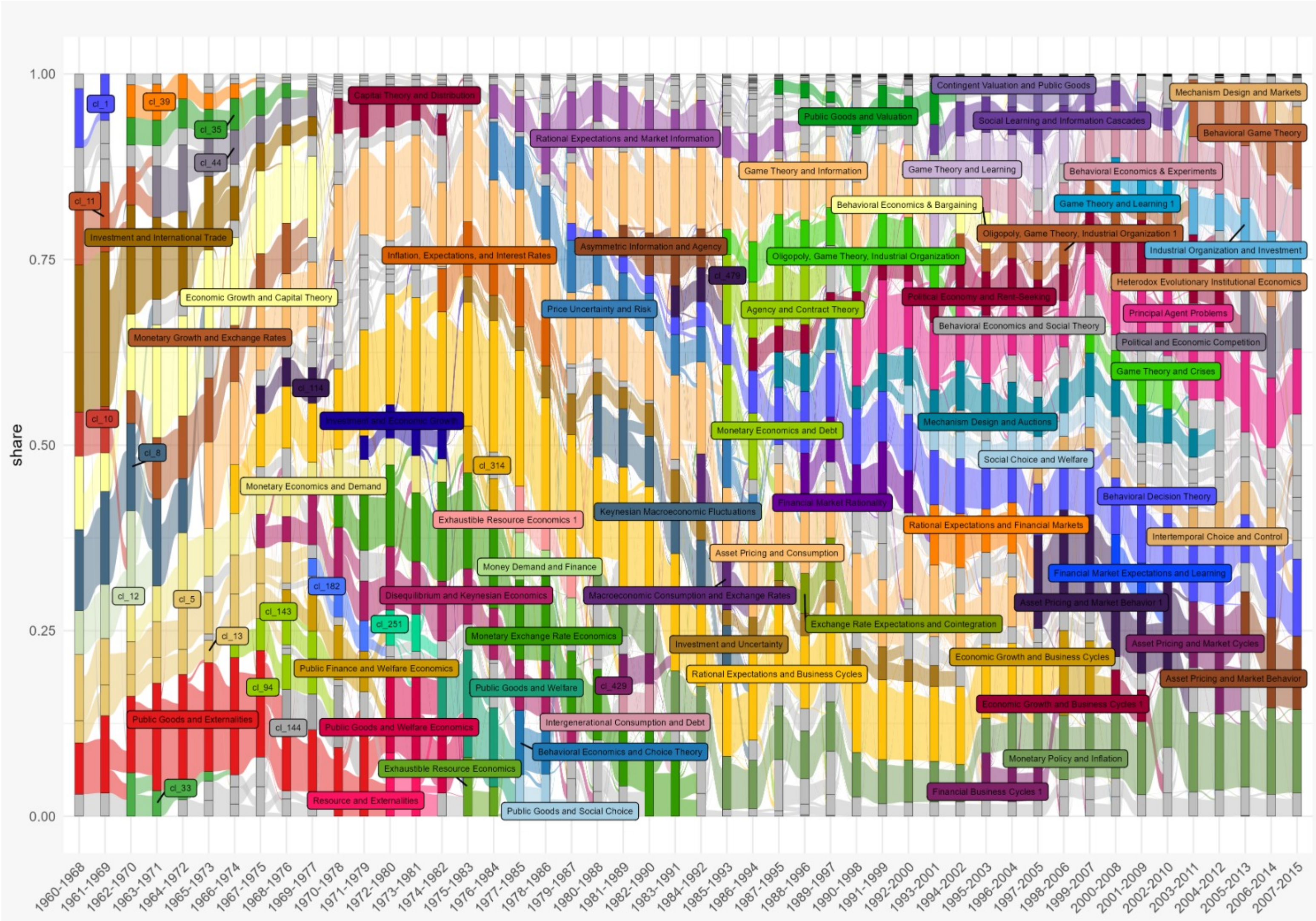


Figure 8.: Alluvial plot of clusters of sentences about rationality

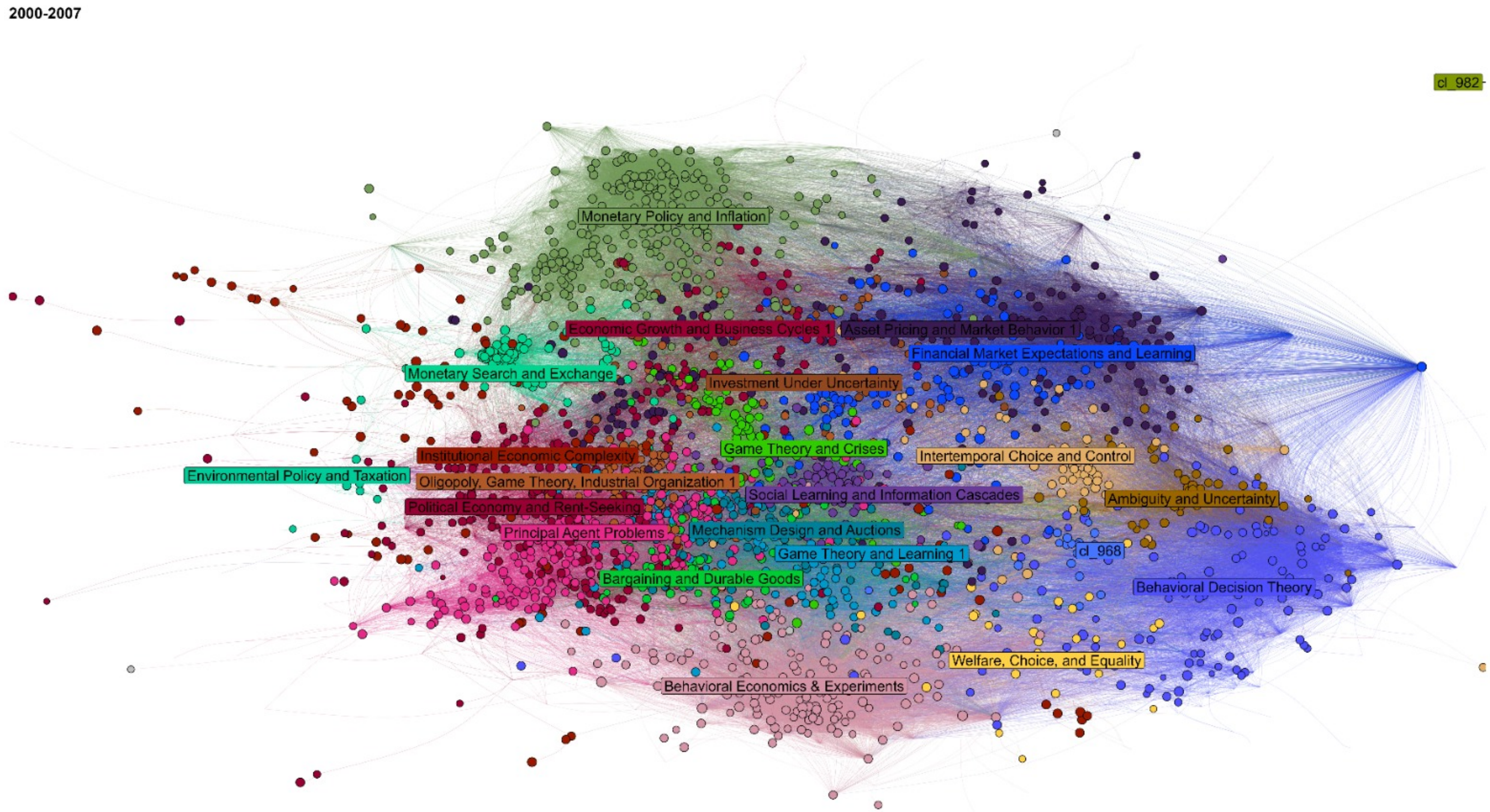


Figure 9.: Example of one bibliographic network representing one horizontal bar of Alluvial (Figure 8)

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